Prodigy Anesthesia Quick Review

Hypoparathyroidism/Hyperparathyroidism

Where are the parathyroid glands located?	There are normally four parathyroid glands located behind the thyroid gland. There are two superior glands and two inferior glands. They are typically around 6 mm long and about 3 millimeters wide. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 962.*
What are the two types of cells found in the parathyroid glands?	The two major types of cells are the chief cells, which secrete parathyroid hormone, and oxyphil cells. The function of oxyphil cells is not known, but it is believed that they may be depleted chief cells that no longer secrete the hormone. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 963.*
What is the principal function of the parathyroid glands?	The parathyroid glands release parathyroid hormone which regulates calcium balance. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 736.*
How does parathyroid hormone increase serum calcium levels?	Parathyroid hormone increases serum calcium levels by increasing bone resorption of calcium, limiting its renal excretion, and enhancing the gastrointestinal absorption of calcium by regulating vitamin D metabolism. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 736.*
How does parathyroid hormone decrease serum phosphate?	By increasing the renal excretion of phosphate. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 736.*
What types of factors would increase the release of parathyroid hormone?	Any condition that results in even a slight decrease in calcium ion concentration in the extracellular fluid will stimulate the release of parathyroid hormone. The parathyroid glands become hyperactive and hypertrophied in conditions such as rickets, pregnancy, and lactation. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 965.*
What are some conditions that would cause the parathyroid glands to decrease the release of parathyroid hormone?	Any condition that would increase the calcium ion concentration in the extracellular fluid such as increased dietary consumption of calcium, increased vitamin D consumption, or the increased release of calcium from the bones from lack of use would decrease the release of parathyroid hormone. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 965.*

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How do the parathyroid glands increase the resorption of calcium and phosphate from the bone?	Parathyroid hormone increases bone resorption of calcium and phosphate in two stages: the first stage begins within minutes and increases the resorption activity of osteoclasts. The second stage is much slower and may require several days or weeks as it stimulates the increased production of osteoclasts. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 963.*
How does parathyroid hormone act on the kidneys?	Parathyroid hormone can cause a rapid loss of phosphate ions in the urine by its effect on the proximal tubule. As more phosphate is excreted, calcium is retained. The increased calcium reabsorption takes place primarily in the collecting tubules and the late distal tubules. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 964.*
What are the cardiac signs and symptoms of hyperparathyroidism?	Hypertension, prolonged PR interval, and a shortened QT interval. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 401.*
What are the neuromuscular signs and symptoms of hyperparathyroidism?	Skeletal muscle weakness, bone demineralization, vertebral collapse, and pathologic fractures. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 400-401.*
What are the renal signs and symptoms of hyperparathyroidism?	Polyuria, polydipsia, decreased glomerular filtration rate, hypophosphatemia, hyperchloremic acidosis and renal stones. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 400-401.*
What is the primary treatment of hyperparathyroidism?	Besides the removal of the diseased portion of the parathyroid glands, treatment of hyperparathyroidism is identical to the treatment for symptomatic hypercalcemia. Saline infusion (typically about 150 mL/hour) is the basic treatment, but the effect of saline to lower calcium is limited and may require the addition of loop diuretics which inhibit both sodium and calcium reabsorption in the proximal loop of Henle. In the case of life-threatening hypercalcemia, bisphosphonates such as disodium etidronate are the drugs of choice. Dialysis may also be used to reduce calcium levels. Calcitonin administration may lower calcium levels, but the effects of calcitonin are short-lived. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 400-401.*
What is secondary hyperparathyroidism?	Secondary hyperparathyroidism is defined as a normal, compensatory increase in parathyroid hormone secretion in response to a disease process or condition that produces hypocalcemia, such as the increased parathyroid hormone secretion associated with chronic renal disease. Because it is a compensatory mechanism, it rarely produces hypercalcemia. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 402.*

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What are the causes of primary hyperparathyroidism?	Primary hyperparathyroidism refers to the excessive secretion of parathyroid hormone from a benign parathyroid tumor, a cancerous parathyroid tumor, or parathyroid hyperplasia. 90% of cases of primary hyperparathyroidism are due to a benign tumor. Incidentally, hyperparathyroidism due to adenoma or hyperplasia is most commonly associated with multiple endocrine neoplasia. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 400.*
How is the care of the parturient with hyperparathyroidism different from that of a nonpregnant patient?	Maternal hypercalcemia can lead to fetal and newborn hypocalcemia and is associated with increased neonatal morbidity and mortality rates. Because the definitive treatment for symptomatic hyperparathyroidism is removal of the parathyroid glands, it is not unusual for a pregnant patient to undergo this surgery. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 154.*
How should you adjust your anesthetic for the patient with primary hyperparathyroidism?	Your anesthetic should be aimed at dealing with underlying hypercalcemia. Because of this, hydration with normal saline and monitoring of urinary output is essential. Because hypercalcemia is associated with somnolence, the anesthetic requirement may be decreased. If personality changes due to chronic hypercalcemia are present, then ketamine may need to be avoided. Baseline skeletal weakness may necessitate a decreased dose of nondepolarizing muscle relaxants, however the increased calcium can antagonize muscle relaxants--in short, hyperparathyroidism is associated with an increased sensitivity to succinylcholine and a resistance to nondepolarizing muscle relaxants. Acidosis increases the serum calcium level, so hypoventilation should be avoided. As with thyroidectomy, there is a risk of damage to the recurrent laryngeal nerve during surgery, so a Nim(Registered) tube or similar device should be used to monitor nerve function during surgery. It is important to position patients with a risk of pathologic fractures carefully. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 401-402.*
What are the signs and symptoms of hypoparathyroidism?	The signs and symptoms of hypoparathyroidism depend on the speed at which the hypocalcemia results. If it is acute, as in inadvertent removal of the parathyroid glands during a thyroidectomy, then the patient may complain of circumoral paresthesias, agitation, and neuromuscular irritability (positive Chvostek's sign). Symptoms due to chronic hypocalcemia include skeletal muscle weakness, prolonged QT interval, calcification of the basal ganglia, and thickening of the skull. The most common cause of chronic hypocalcemia is chronic renal failure. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 402.*

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What is the difference between hypoparathyroidism and pseudohypoparathyroidism?	Hypoparathyroidism occurs when the secretion of parathyroid hormone is decreased or absent or the tissues do not respond to the released hormone. The most common cause of decreased or absent parathyroid hormone is the removal of the parathyroid glands during thyroidectomy. Pseudohypoparathyroidism occurs when the hormone is released normally, but the kidneys are unable to respond to the hormone. Pseudohypoparathyroidism is associated with mental retardation, basal ganglia calcification, obesity, decreased height, and short metacarpals and metatarsal bones. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 402.*
What is the treatment for hypoparathyroidism?	The treatment of hypoparathyroidism is aimed at restoring low calcium levels to normal. An infusion of 10 mL of calcium gluconate 10% IV should be administered until signs of neuromuscular irritability resolve. Respiratory or metabolic alkalosis should be normalized. Thiazide diuretics may be helpful as they result in sodium depletion without the loss of potassium which tends to increase calcium levels. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 402-403.*
What are the anesthetic considerations for the patient with hypoparathyroidism?	The anesthetic implications are the same as those for a patient with hypocalcemia. It is important to avoid hyperventilation as the resulting alkalosis can decrease ionized calcium concentrations. Administering whole blood with citrate does not affect calcium concentrations because calcium can be mobilized quickly into the bloodstream, however, the rapid administration of blood can result in loss of ionized calcium. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 403.*

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